Q1 - 25 July - Shift 1

A line, with the slope greater than one, passes through the point A(4, 3) and intersects the line x – y – 2 = 0 at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line :

(A) $2x + y = 9$ (B) $3x - 2y = 7$
(C) $x + 2y = 6$ (D) $2x - 3y = 3$

Space for your notes:

Q2 - 25 July - Shift 2

Let the point P (α, β) be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then 100 (α + β) is equal to

(A) –14 (B) 42
(C) –22 (D) 14

Space for your notes:

Q3 - 26 July - Shift 1

A point P moves so that the sum of squares of its distances from the points (1, 2) and (–2, 1) is 14. Let f(x, y) = 0 be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the point C, D . Then the area of the quadrilateral ACBD is equal to

(A) $\frac{9}{2}$ (B) $\frac{3\sqrt{17}}{2}$
(C) $\frac{3\sqrt{17}}{4}$ (D) 9

Space for your notes:

Q4 - 26 July - Shift 1

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is $(2, a)$, $-\frac{1}{2} < a < 2$, then p is equal to

Space for your notes:

Q5 - 27 July - Shift 1

Let A(1, 1), B(-4, 3) C(-2, -5) be vertices of a triangle ABC, P be a point on side BC, and Δ_1 and Δ_2 be the areas of triangle APB and ABC. Respectively.

If $\Delta_1 : \Delta_2 = 4 : 7$, then the area enclosed by the lines AP, AC and the x-axis is

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$

(C) $\frac{1}{2}$ (D) 1

Space for your notes:

Q6 - 27 July - Shift 2

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 39$ and $x - y = 3$ respectively and P(2, 3) is its circumcentre. Then which of the following is NOT true :

(A) $(AC)^2 = 9p$

(B) $(AC)^2 + p^2 = 136$

(C) $32 < \text{area}(\Delta ABC) < 36$

(D) $34 < \text{area}(\Delta ABC) < 38$

Space for your notes:

Q7 - 28 July - Shift 1

For $t \in (0, 2\pi)$, if ABC is an equilateral triangle with vertices A(sint, –cost), B(cost, sint) and C(a, b) such that its orthocentre lies on a circle with centre $\left(1, \frac{1}{3}\right)$, then $(a^2 - b^2)$ is equal to :

(A) $\frac{8}{3}$ (B) 8

(C) $\frac{77}{9}$ (D) $\frac{80}{9}$

Space for your notes:

Q8 - 29 July - Shift 1

Let the circumcentre of a triangle with vertices A(a, 3), B(b, 5) and C(a, b), ab > 0 be P(1, 1). If the line AP intersects the line BC at the point $Q(k_1, k_2)$, then $k_1 + k_2$ is equal to :

(A) 2 (B) $\frac{4}{7}$ (C) $\frac{2}{7}$ (D) 4

Space for your notes:

Q9 - 29 July - Shift 2

Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3\left(m_2^2 + m_2^2\right) = 220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)x + (\sin\alpha + \cos\alpha)y = 10$, then $72(\sin^4\alpha + \cos^4\alpha) + a^2 - 3a + 13$ is equal to:

(A) 119 (B) 128

(C) 145 (D) 155

Space for your notes:

Q10 - 29 July - Shift 2

Let A $(\alpha, -2)$, B $(\alpha, 6)$ and C$\left(\frac{\alpha}{4}, -2\right)$ be vertices of a ΔABC. If $\left(5, \frac{\alpha}{4}\right)$ is the circumcentre of ΔABC, then which of the following is NOT correct about ΔABC:

(A) ares is 24 (B) perimeter is 25

(C) circumradius is 5 (D) inradius is 2

Space for your notes:

Answer Key

Q1 (C)	**Q2** (D)	**Q3** (B)	**Q4** (3)
Q5 (C)	**Q6** (D)	**Q7** (B)	**Q8** (B)
Q9 (B)	**Q10** (B)		

Q1 (C)

Let $B(x_1, x_1 - 2)$

$$\sqrt{(x_1 - 4)^2 + (x_1 - 2 - 3)^2} = \frac{\sqrt{29}}{3}$$

Squaring on both side

$$18x_1^{\,2} - 162x_1 + 340 = 0$$

$$x_1 = \frac{51}{9} \quad \text{or} \quad x_1 = \frac{10}{3}$$

$$y_1 = \frac{33}{9} \quad \text{or} \quad y_1 = \frac{4}{3}$$

Option (C) will satisfy $\left(\frac{10}{3}, \frac{4}{3}\right)$

Q2 (D)

(D)

Q3 (B)

$(x-1)^2+(y-2)^2+(x+2)^2+(y-1)^2=14$

$\Rightarrow x^2+y^2+x-3y-2=0$

Put $x=0$

$\Rightarrow y^2-3y-2=0$

$\Rightarrow y=\frac{3\pm\sqrt{17}}{2}$

Put $y=0$

$\Rightarrow x^2+x-2=0$

$(x+2)(x-1)=0$

$\therefore A(-2,0), B(1,0), C\left(0,\frac{3+\sqrt{17}}{2}\right), D\left(0,\frac{3-\sqrt{17}}{2}\right)$

Area $=\frac{1}{2}.3.\sqrt{17}=\frac{3\sqrt{17}}{2}$

Q4 (3)

Coordinates of $A(1,-2)$, $B\left(\frac{15a}{1-2p},\frac{-30a}{1-2p}\right)$ and

orthocentre $H(2, a)$

Slope of AH = p

$a+2=p$ .....(1)

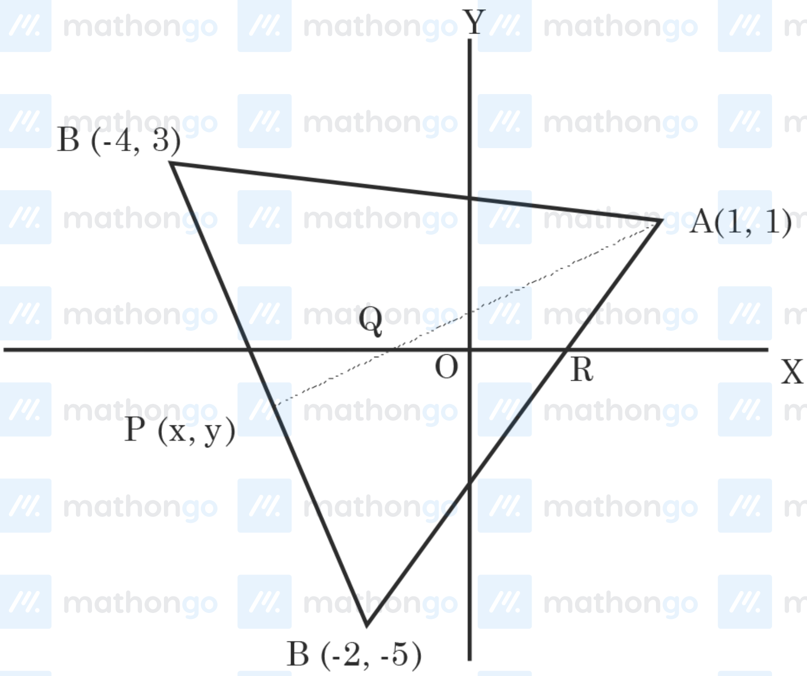
Slope of BH = –1

$31a-2ab=15a+4p-2$ .....(2)

From (1) and (2)

$a=1$ & $p=3$

Q5 (C)



Given $\Delta_1 = \frac{1}{2}\begin{vmatrix} x & y & 1 \\ 1 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$

& $\Delta_2 = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ -4 & 3 & 1 \\ -2 & -5 & 1 \end{vmatrix}$

Given $\frac{\Delta_1}{\Delta_2} = \frac{4}{7} \Rightarrow \frac{-2x-5y+7}{36} = \frac{4}{7}$

$\Rightarrow 14x + 35y = -95$ ....(1)

Equation of BC is 4x + y = -13 ....(2)

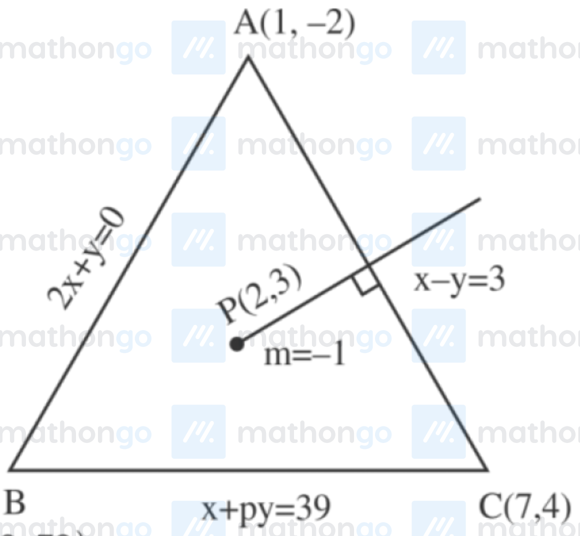
Solve equation (1) & (2)

Point $P\left(\frac{-20}{7}, \frac{-11}{7}\right)$

Here point $Q\left(\frac{-1}{2}, 0\right)$ & $R\left(\frac{1}{2}, 0\right)$

So Area of triangle AQR $= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

Q6 (D)



$\left(\frac{-39}{15}, \frac{78}{15}\right)$

Perpendicular bisector of AB

$x + y = 5$

Take image of A

$$\frac{x-1}{1} = \frac{y+2}{1} = \frac{-2(-6)}{2} = 6$$

(7, 4)

$7 + 4p = 39$

$p = 8$

solving $x + 8y = 39$ and $y = -2x$

$x = \frac{-39}{15} \qquad y = \frac{78}{15}$

$AC^2 = 72 = 9p$

$AC^2 + p^2 = 72 + 64 = 136$

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} 1 & -2 & 1 \\ 7 & 4 & 1 \\ \frac{-39}{15} & \frac{78}{15} & 1 \end{vmatrix}$$

$= \frac{1}{2}\left[4 - \frac{78}{\cdots} + 2\left(7 + \frac{39}{\cdots}\right) + 7\left(\frac{78}{\cdots}\right) + \frac{4\times 39}{\cdots}\right]$

$$= \frac{1}{2}\left[18+18\times\frac{13}{5}\right]$$

$$= 9\left[\frac{18}{5}\right]=\frac{162}{5} = 32.4$$

Ans. (D)

Q7 (B)

$s \equiv \sin t, c \equiv \cos t$

Let orthocentre be (h,k)

Since it if an equilateral triangle hence orthocentre coincides with centroid.

$\therefore\ a+s+c=3h, b+s-c=3k$

$\therefore\ (3h-a)^2+(3k-b)^2=(s+c)^2+(s-c)^2=2\left(s^2+c^2\right)=2$

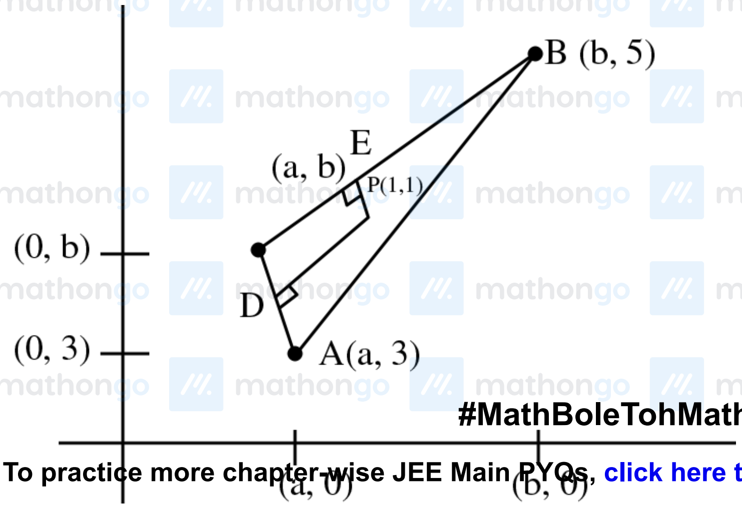
$\therefore\ \left(h-\frac{a}{3}\right)^2+\left(K-\frac{b}{3}\right)^2=\frac{2}{9},$

circle centre at $\left(\frac{a}{3},\frac{b}{3}\right)$

Gives, $\frac{a}{3}=1,\frac{b}{3}=\frac{1}{3}\Rightarrow a=3, b=1$

$\therefore\ a^2-b^2=8$

Q8 (B)



$m_{AC} \longrightarrow \infty$

$m_{PD} = 0$

$D\left(\frac{a+a}{2}, \frac{b+3}{2}\right)$

$D\left(a, \frac{b+3}{2}\right)$

$m_{PD} = 0$

$\frac{b+3}{2} - 1 = 0$

$b + 3 - 2 = 0$

$\boxed{b = -1}$

$E\left(\frac{b+a}{2}, \frac{5+b}{2}\right) = \left(\frac{af}{2}, 2\right)$

$m_{CB} \,.\, m_{EP} = -1$

$\left(\frac{5-b}{b-a}\right) = \left(\frac{2-1}{\frac{a-1}{2} - 1}\right) = -1$

$\left(\frac{6}{-1-a}\right) = \left(\frac{2}{a-3}\right) = -1$

$12 = (1 + a)(a - 3)$

$12 = a^2 - 3a + a - 3$

$\Rightarrow a^2 - 2a - 15 = 0$

$(a - 5)(a + 3) = 0$

$a = 5$ or $a = -3$

Given $ab > 0$

$a(-1) > 0$

$a > 0$

$a < 0$

$\boxed{a = -3}$ Accept

AP line A (–3, 3) P(1, 1)

$$y - 1 = \left(\frac{3-1}{-3-1}\right)(x-1)$$

$-2y + 2 = x - 1$

$\Rightarrow \boxed{x + 2y = 3}$ Appling .....(1)

Line BC B(–1, 5)

C(–3, –1)

$$(y-5) = \frac{6}{2}(x+1)$$

$y - 5 = 3x + 3$

$\boxed{y = 3x + 8}$ .....(2)

Solving (1) & (2)

$x + 2(3x + 8) = 3$

$\Rightarrow 7x + 16 = 3$

$7x = -13$

$$x = -\frac{13}{7}$$

$$y = 3\left(-\frac{13}{7}\right) + 8$$

$$= \frac{-39+56}{7}$$

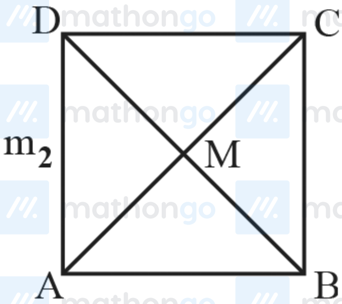
$$y = \frac{17}{7}$$

$$x + y = \frac{-13+17}{7} = \frac{4}{7}$$

Q9 (B)

$$m_1 m_2 = -1$$

$$a^2 + 11a + 3\left(m_1^2 + \frac{1}{m_1^2}\right) = 220$$



Eq. of AC

$$AC = (\cos\alpha - \sin\alpha) + (\sin\alpha + \cos\alpha)\, y = 10$$

$$BD = (\sin\alpha - \cos\alpha)\, x + (\sin\alpha - \cos\alpha)\, y = 0$$

$$(10\,(\cos\alpha - \sin\alpha),\ 10\,(\sin\alpha - \cos\alpha))$$

$$\text{Slope of AC} = \left(\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right) = \tan\theta = M$$

Eq. of line making an angle $^{\pi}/_{4}$ with AC

$$m_1, m_2 = \frac{m \pm \tan\frac{\pi}{4}}{1 \pm m\tan\frac{\pi}{4}}$$

$$= \frac{m+1}{1-m} \text{ or } \frac{m-1}{1+m}$$

$$\frac{\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} + 1}{1 - \left(\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}\right)}, \frac{\frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} - 1}{1 + \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}}$$

$$m_1, m_2 = \tan\alpha\ ,\ \cot\alpha$$

mid point of AC & BD

$= M\,(5(\cos\alpha - \sin\alpha),\ 5(\cos\alpha + \sin\alpha))$

$B\,(10(\cos\alpha - \sin\alpha),\ 10(\cos\alpha + \sin\alpha))$

$$a = AB = \sqrt{2}\,BM = \sqrt{2}\left(5\sqrt{2}\right) = 10$$

$a = 10$

$$\because a^2 + 11a + 3\left(m_1^2 + \frac{1}{m_1^2}\right) = 220$$

$100 + 110 + 3\,(\tan^2\alpha + \cot^2\alpha) = 220$

$$\text{Hence } \tan^2\alpha = 3,\ \tan^2\alpha = \frac{1}{3} \Rightarrow \alpha = \frac{\pi}{3} \text{ or } \frac{\pi}{6}$$

Now $72\,(\sin^4\alpha + \cos^4\alpha) + a^2 - 3a + 13$

$$= 72\left(\frac{9}{16} + \frac{1}{16}\right) + 100 - 30 + 13$$

$$= 72\left(\frac{5}{8}\right) + 83 = 45 + 83 = 128$$

Q10 (B)

A $(\alpha, -2)$: B $(\alpha, 6)$: C $\left(\frac{\alpha}{4}, -2\right)$

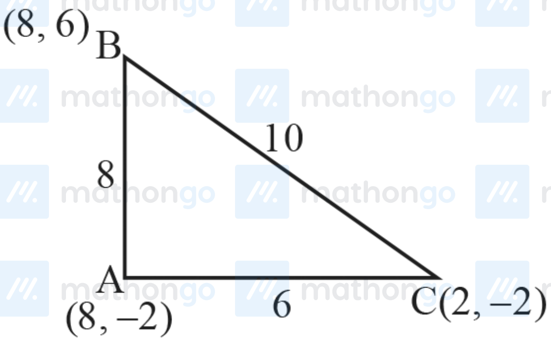
since AC is perpendicular to AB.

So, ΔABC is right angled at A.

Circumcentre = mid point of BC. = $\left(\frac{5\alpha}{8}, 2\right)$

$\therefore \frac{5\alpha}{8} = 5$ & $\frac{\alpha}{4} = 2$

$\alpha = 8$



Area = $\frac{1}{2}(6)(8) = 24$

Perimeter = 24

Circumradius = 5

Inradius = $\frac{\Delta}{s} = \frac{24}{12} = 2$